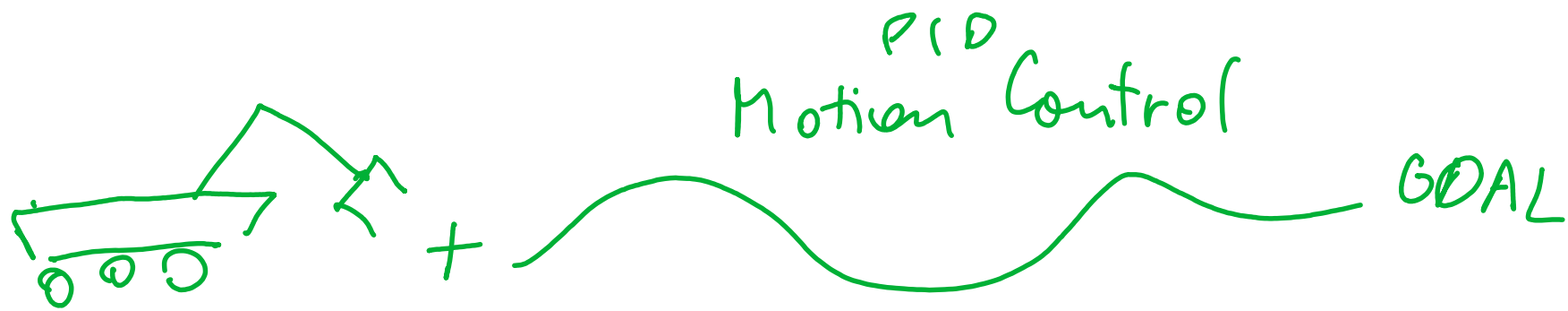




1 – Introduction

Advanced Methods for Mapping and Self-localization in Robotics
MPC-MAP

Petr Gabrlik
Brno University of Technology
2026

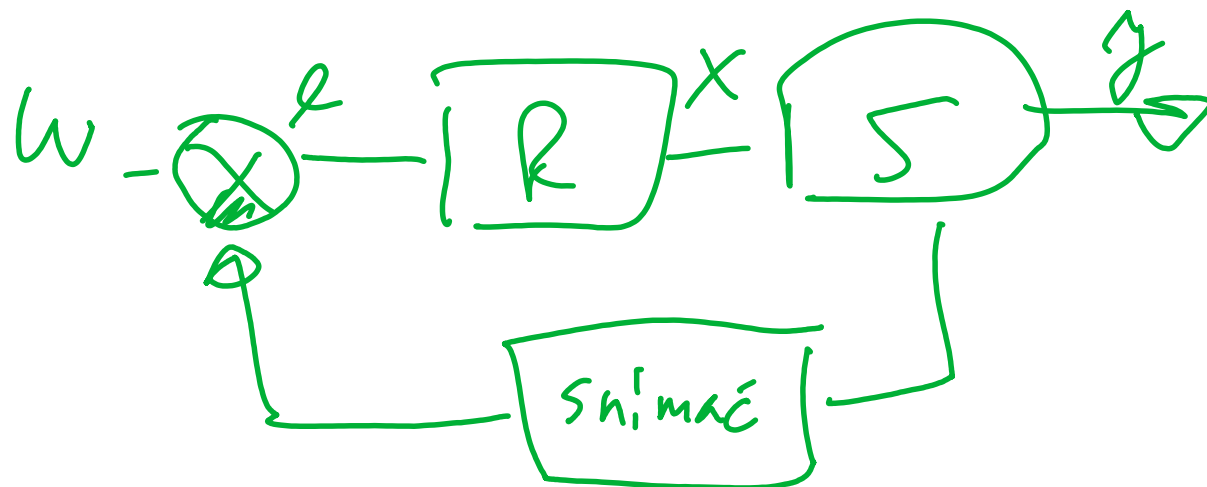


PID
Motion Control

LOCALISATION
- Particle filter
- extended KF

Path Planning

A^* , Dijkstra



Mapping - Static
Dynamic

SCAM

Perception - LIDAR



Sc

Motivation

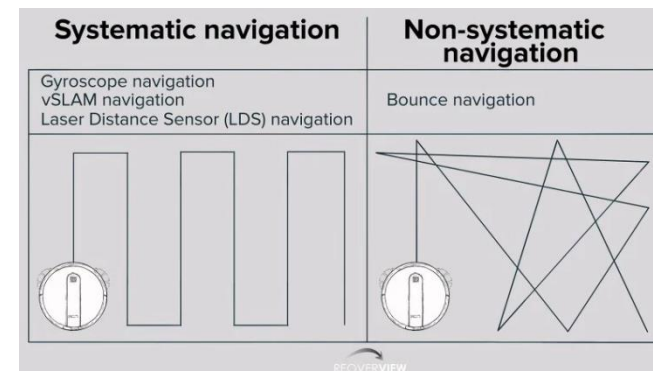
The current use of (autonomous) mobile robots



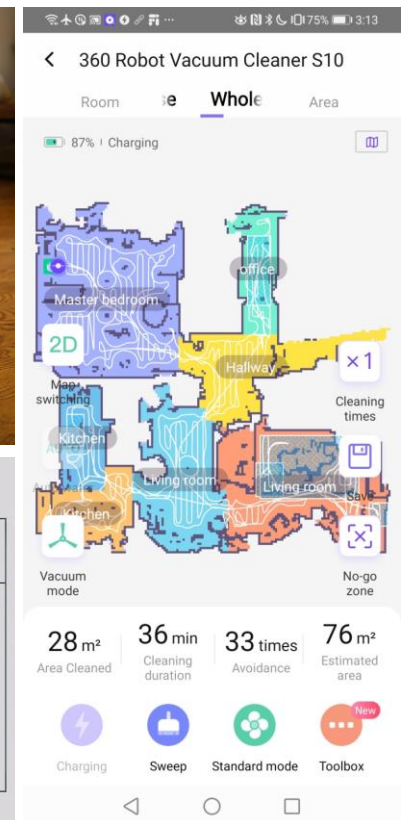
Robot Vacuums

- A widely used type of a consumer-grade robot
- Both *dump* and *high-tech* solutions *bez úprav / prostredia*
- **Environment:** indoor, 2D, uncontrolled
- **Task / navigation goal:** area coverage
- **Localization:** none, encoders, IMU, LiDAR SLAM
- **Perception:** tactile sensors, cameras, LiDAR etc.

Source: iRobot



Source: reoverview.co.uk



Source: digitaltrends.com



Robotic Lawn Mower

- A similar task as a vacuum cleaner
- Both *dump* and *high-tech* solutions
- **Environment:** outdoor, 3D, partially controlled/uncontrolled
- **Task / navigation goal:** area coverage
- **Localization:** none, encoders, IMU, GNSS
- **Perception:** tactile sensors, cameras, LiDAR etc.



Source: wsj.com

Warehouse Robots

- Robotic picker, pallet trucks, goods-to-person
- **Environment:** indoor, 2D, controlled/uncontrolled
- **Task / navigation goal:** go to location
- **Localization:** encoders, IMU, LiDAR-based localization in map, indoor localization systems (MoCap, UWB, fiducial markers etc.) Art Tag
- **Perception:** cameras, depth-cameras, LiDARs etc.



Source: ep-equipment.com



Source: theverge.com



Food Delivery Robots

- A robotic replacement for waiters
- **Environment:** indoor, 2D, uncontrolled
- **Task / navigation goal:** go to location
- **Localization:** encoders, IMU, LiDAR-based localization in map, LiDAR SLAM
- **Perception:** tactile sensors, cameras, depth-cameras, LiDARs etc.

Source: kvados.cz



Source: globaltimes.cn

Cleaning Robots

- Floor cleaning, UV disinfection
- **Environment:** indoor, 2D, uncontrolled
- **Task / navigation goal:** area coverage, go to location
- **Localization:** encoders, IMU, LiDAR-based localization in map, LiDAR SLAM
- **Perception:** tactile sensors, cameras, depth-cameras, LiDARs etc.

Star ma trhu - este nie moc
vyniz+elne medzi ludmi

Source: cbc.ca



Source: healthcareitnews.com



RTK GNSS - přesnost až mm

Agricultural robots

- Seeding, harvesting, crop maintenance etc.
- **Environment:** outdoor/indoor, 3D, partially controlled/uncontrolled
- **Task / navigation goal:** go to location, area coverage, linear trajectories
- **Localization:** encoders, IMU, GNSS, indoor localization systems, LiDAR-based localization in map, LiDAR SLAM
- **Perception:** cameras, depth-cameras, LiDARs

ag. tab.
dôležité

Uncontrolled - menia sa podmienky,
prostredie

Source: idnes.cz



Source: latimes.com



Last-Mile Delivery Robots

- Delivering goods to customers
- Operates mainly on the pavements
- **Environment:** outdoor, 3D, uncontrolled
- **Task / navigation goal:** go to location
- **Localization:** encoders, IMU, GNSS
- **Perception:** cameras, depth-cameras, LiDARs etc.



Source: freightwaves.com



Source: cambridgeindependent.co.uk



Self-Driving Cars

- Comprehensive task with human health risks
- **Environment:** outdoor, 3D, uncontrolled
- **Task / navigation goal:** go to location
- **Localization:** encoders, IMU, GNSS
- **Perception:** cameras, depth-cameras, LiDARs, radars

Source: marketwatch.com

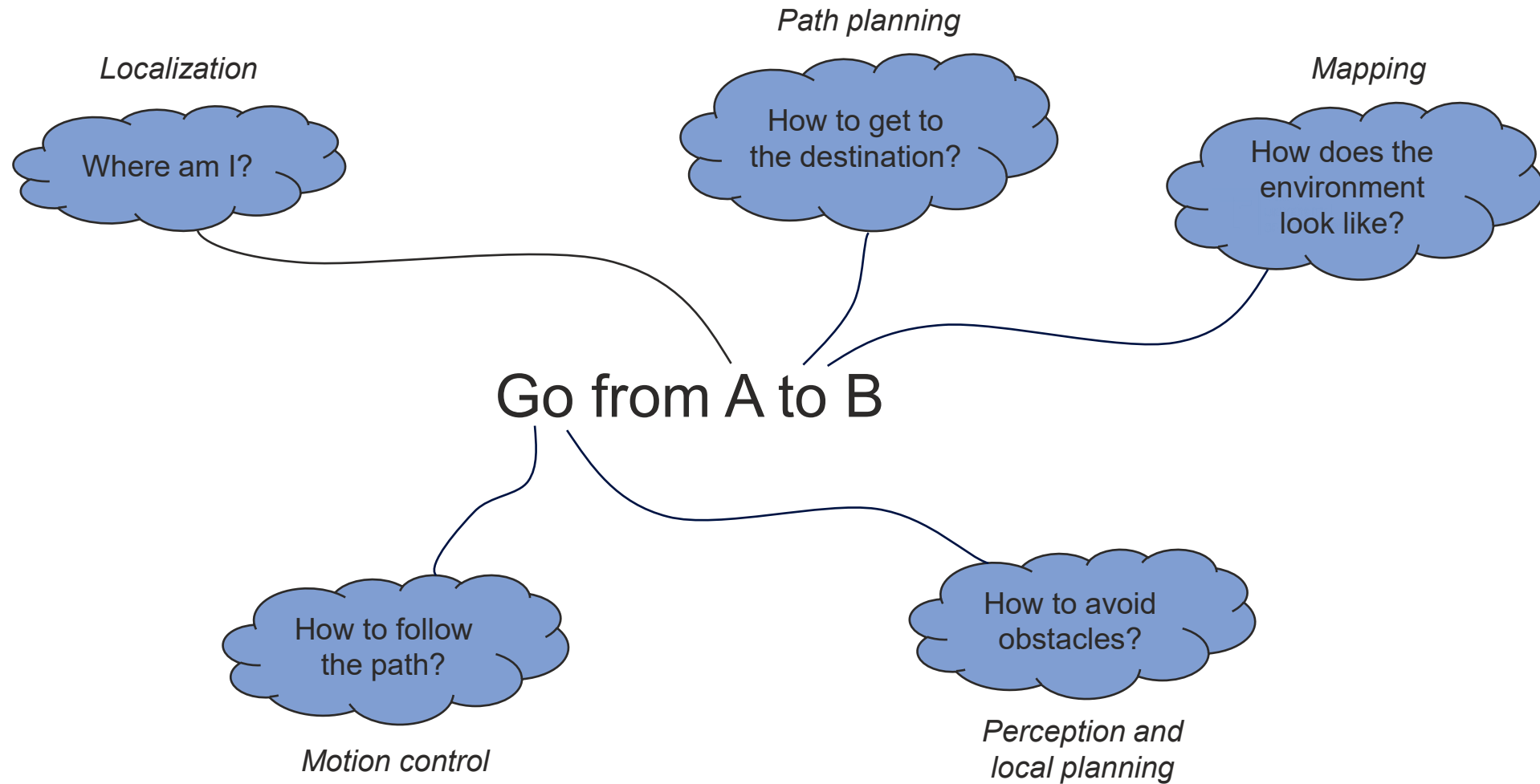


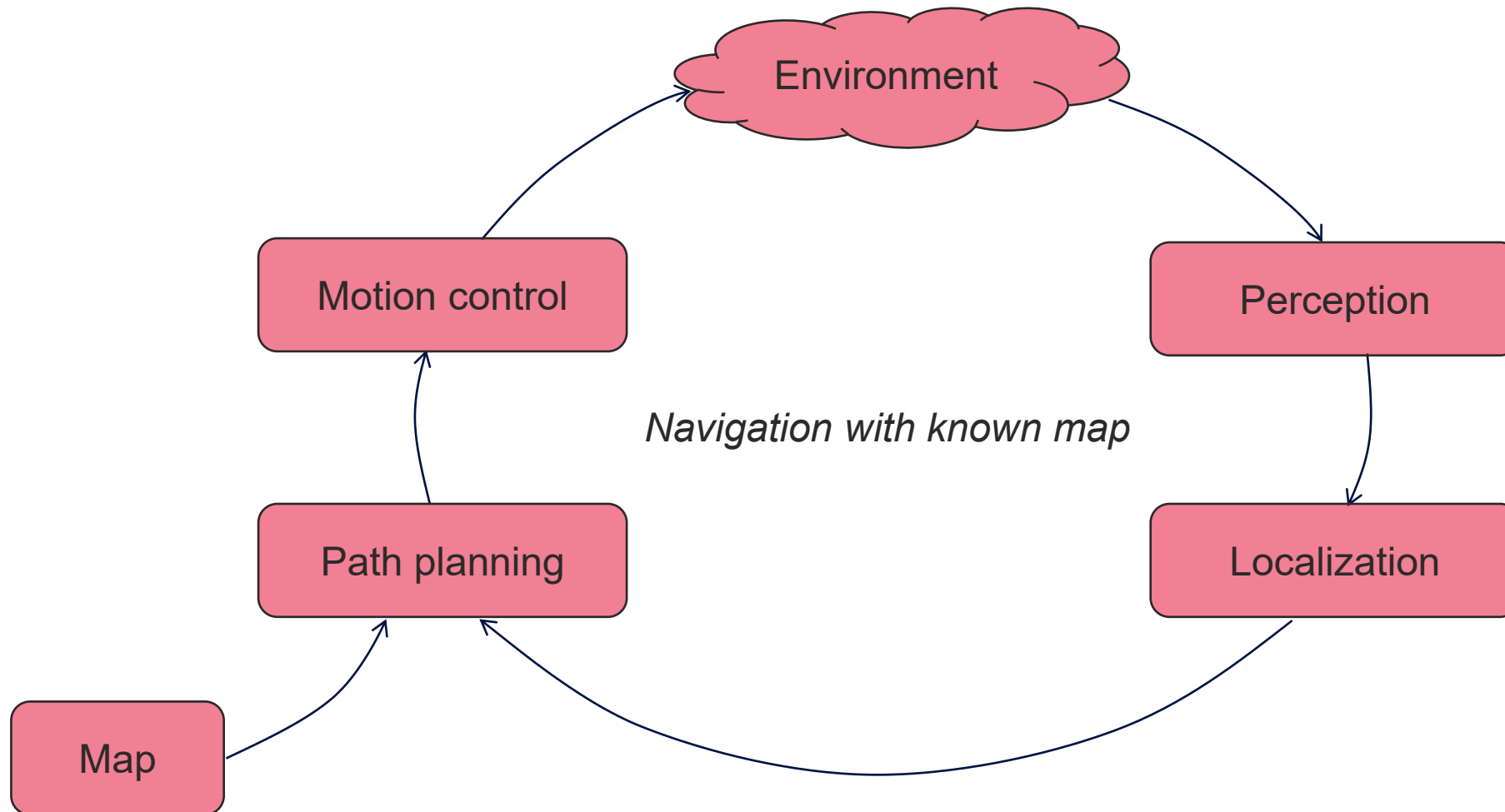
Source: waymo.com

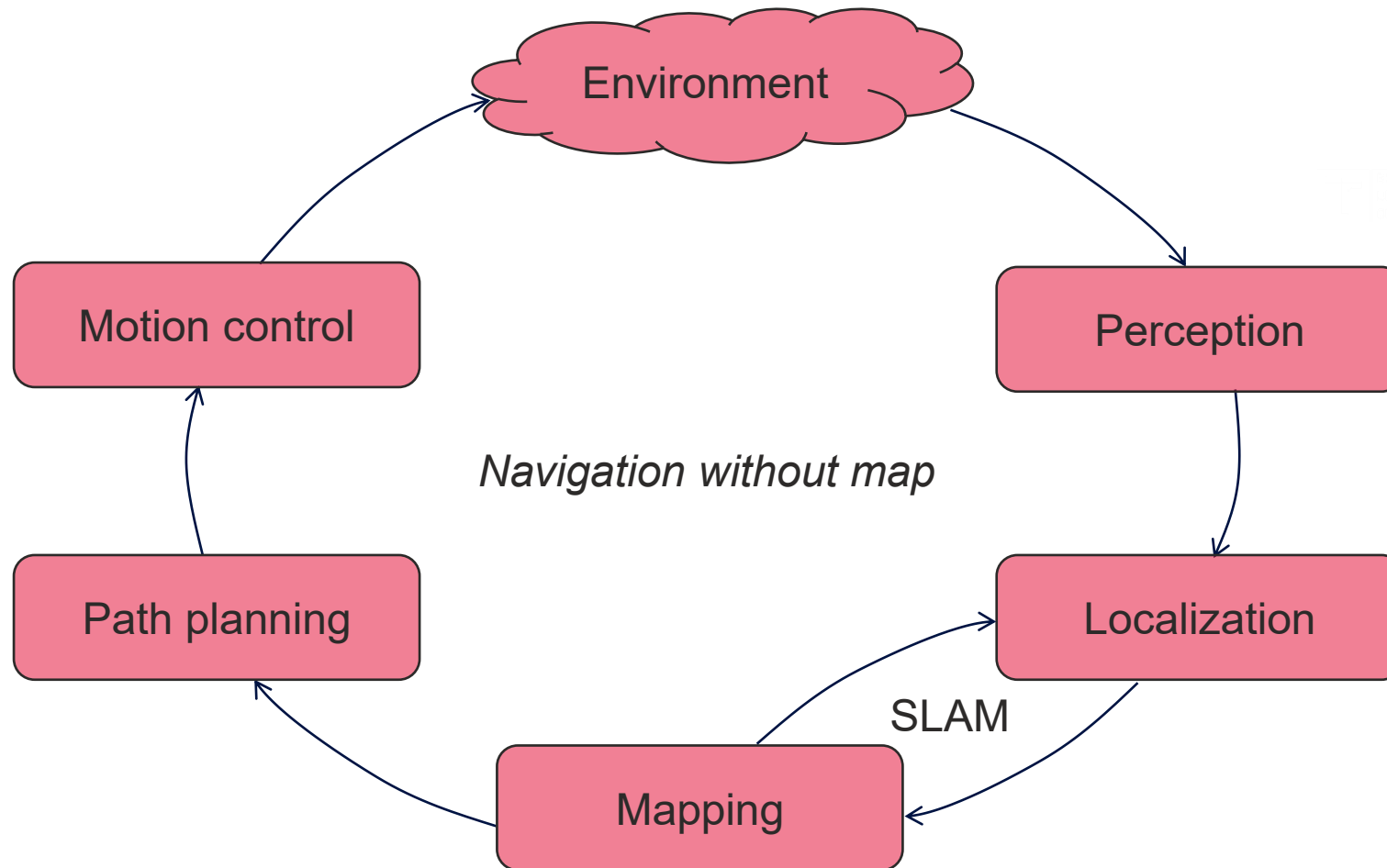


Problem Specification

Fundamental problems in autonomous mobile robotics









Course Details

The goals, lecturers, structure, resources



▪ Motivation and goals

- Diving into the fundamental problems of mobile robotics.
- Understand robot navigation: self-localization, path planning, motion control and map building topics.
- Algorithm-focused course (no complex SW, no HW/robot, no communication..).
- Practice theory via practical tasks in the simulator.

▪ Course parameters

- 3 credits (ECTS), ended with classified credit
- Total work: **72 hrs.** (~24 hrs./credit)
- Lectures: 7×2 hrs. (optional, highly recommended)
- Labs: 7×2 hrs., weekly assignments – individual homework with consultations during lab hours (compulsory)
- Total individual homework: **58 hrs.**, ~ **9 hrs./week** (weekly projects, semestral project, self-study)



- **Course supervisor**

- Ing. Petr Gábrlík, Ph.D.

- **Lecturers**

Ing. Petr Gábrlík, Ph.D.
gabrlík@vut.cz
SE1. 112



Ing. Tomáš Lázna, Ph.D.
tomas.lazna@ceitec.vutbr.cz
SE 1.112

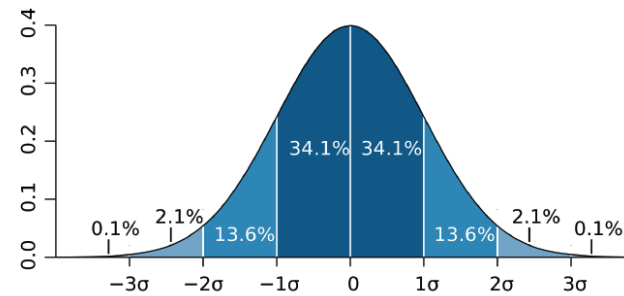




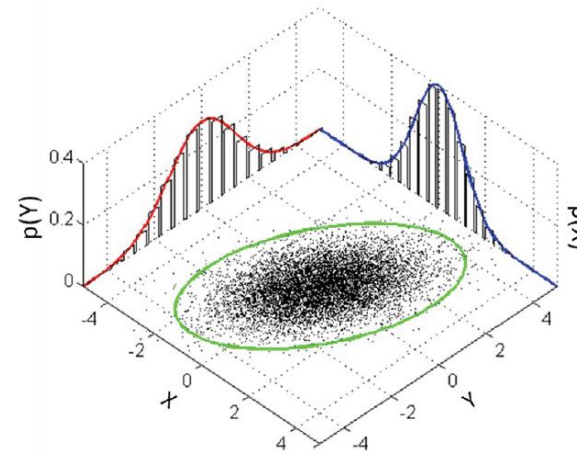
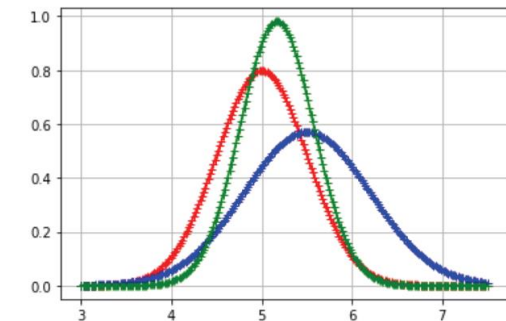
Week	Lecture	Lab	Lab points	Lecturer
1	Introduction	Robot navigation demo	0	Gabrlík
2	Probabilistics Robotics	Uncertainty	10	Gabrlík
3	Kinematics and Motion Control	Motion Control	10	Lazna
4	Particle Filter	Particle Filter	10	Lazna
5	Kalman Filter and EKF	Kalman Filter and EKF	10	Gabrlík
6	Path Planning	Path planning	10	Lazna
7	SLAM	Individual work on project	50	Gabrlík



- Introduction to basic **statistics** and **probability**
- Normal distribution
- Covariance matrix
- Bayes theorem
- Markov localization
- Basic sensors



Source: wikipedia.org



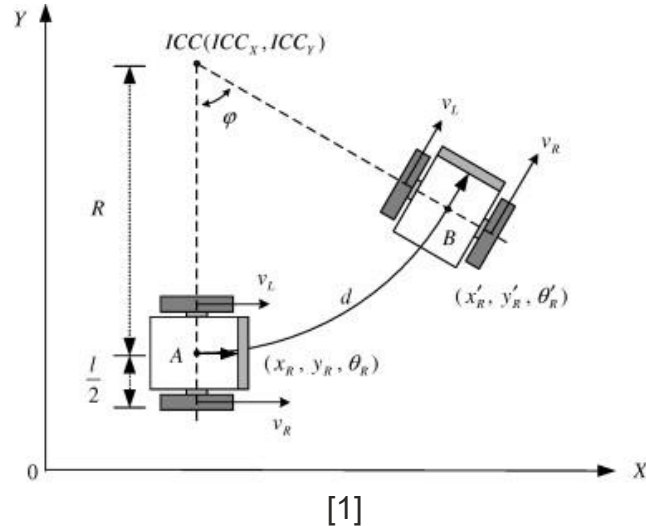
Source: wikipedia.org

$$\text{Cov} = \begin{bmatrix} \text{Cov}(X, X) & \text{Cov}(X, Y) & \text{Cov}(X, Z) \\ \text{Cov}(Y, X) & \text{Cov}(Y, Y) & \text{Cov}(Y, Z) \\ \text{Cov}(Z, X) & \text{Cov}(Z, Y) & \text{Cov}(Z, Z) \end{bmatrix}$$

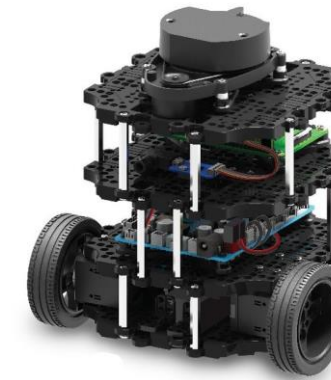
Covariance matrix



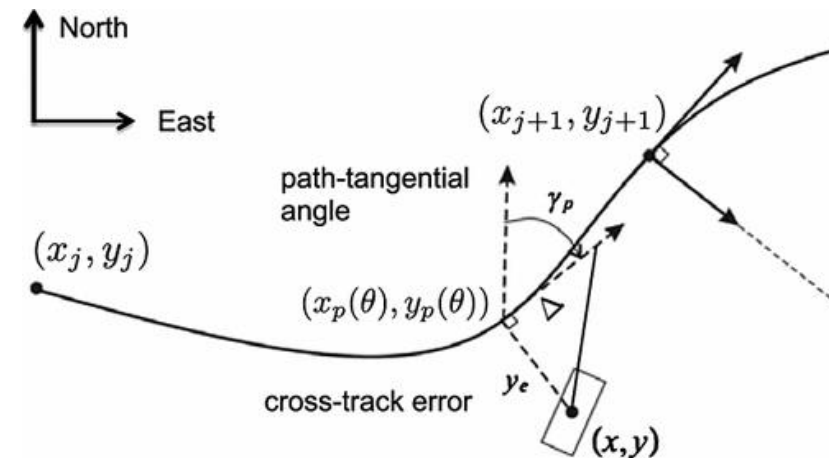
- **Kinematics** of a differential drive
- Motion model
- How to **follow a path**?



[1]



[3]

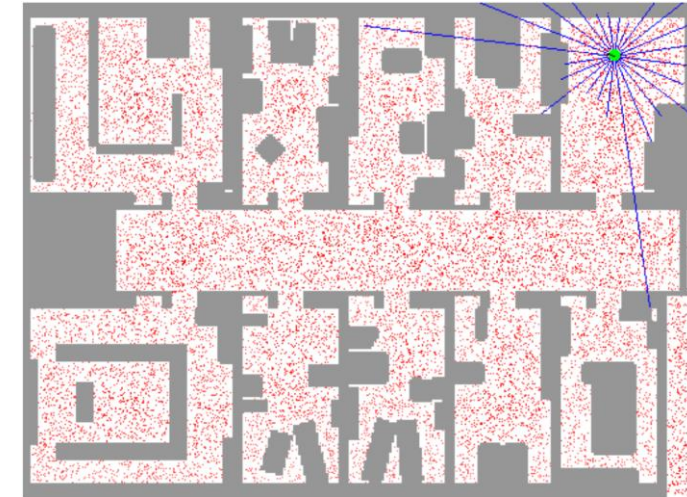


[2]

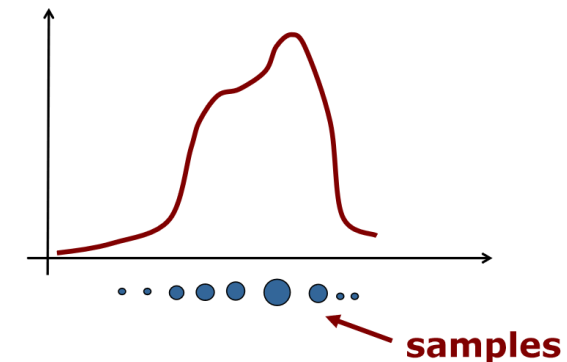
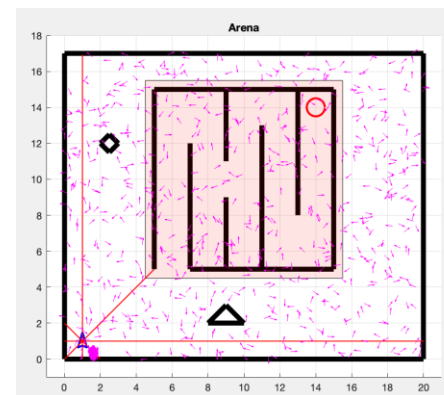
- [1] HAN, Soonshin, ByoungSuk CHOI a JangMyung LEE. A precise curved motion planning for a differential driving mobile robot. *Mechatronics* [online]. 2008, **18**(9), 486-494 [cit. 2022-02-04]. DOI: 10.1016/j.mechatronics.2008.04.001
- [2] FOSSEN, Thor I. a Anastasios M. LEKKAS. Direct and indirect adaptive integral line-of-sight path-following controllers for marine craft exposed to ocean currents. *International Journal of Adaptive Control and Signal Processing* [online]. 2017, **31**(4), 445-463 [cit. 2022-02-04]. DOI: 10.1002/acs.2550
- <https://www.turtlebot.com/>



- Introduction to **Monte Carlo** methods
- What are **particles**?
- Random numbers sampling – a vital element
- Particle filter algorithm – 3 steps:
 - Prediction
 - Correction
 - Resampling
- Application of the PF to the **localization problem**
- Typical issues and how to solve them



[1]



[2]

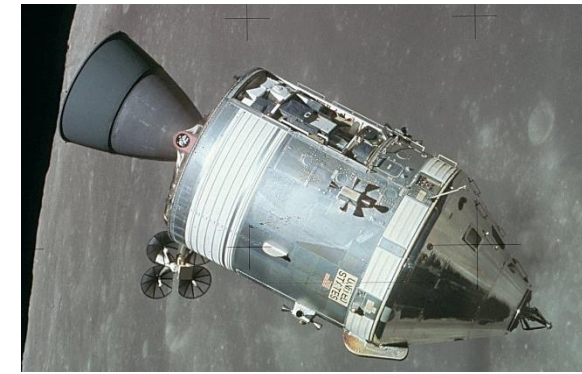
[1] TRIEBEL, Rudolph. The Particle Filter. In: *Machine Learning for Computer Vision* [online]. Technische Universität München, 2017 [cit. 2021-02-19]. Available at: https://vision.in.tum.de/_media/teaching/ss2017/ml4cv/variationalinference.pdf

[2] STACHNISS, Cyrill. Short Introduction to Particle Filters and Monte Carlo Localization [online]. Uni Freiburg, 2013 [cit. 2021-02-18]. Available at: <http://ais.informatik.uni-freiburg.de/teaching/ws13/mapping/pdf/slam10-particle-filter-4.pdf>

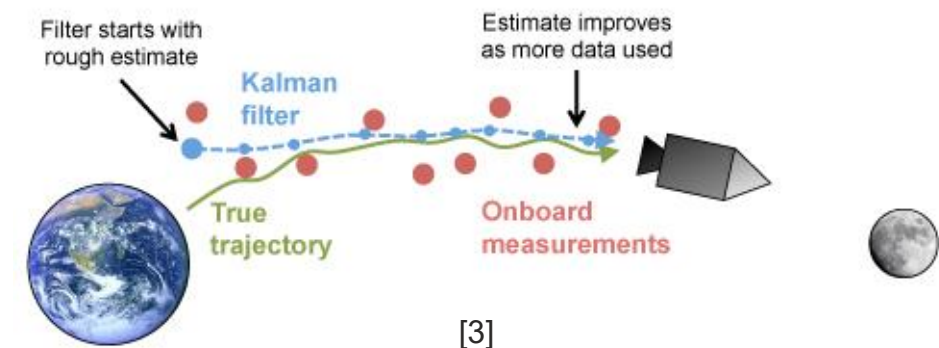


міжна в іпоіст. навігація

- An algorithm for **filtering** and **prediction** in linear systems / **estimating** unknown variables.
- Suitable for fusing data from different sensors (different variables and sampling periods).
- Used for **trajectory estimation** for the **Apollo** program in the ~1960s – one of the very first applications of the Kalman filter [1].
 - Combination of acceleration data and star position observation.



[2]

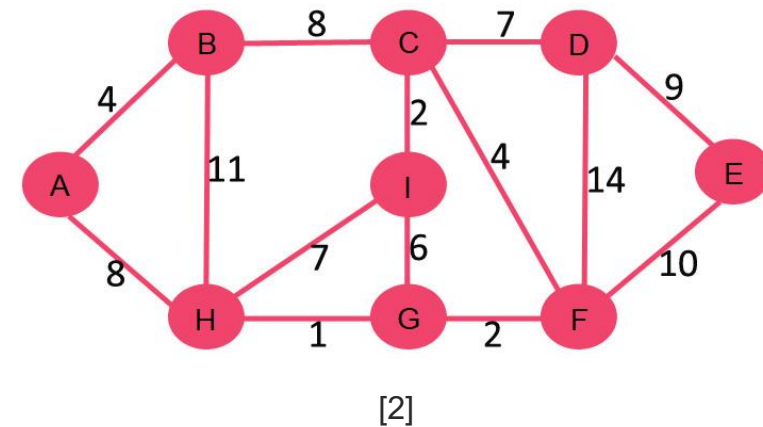
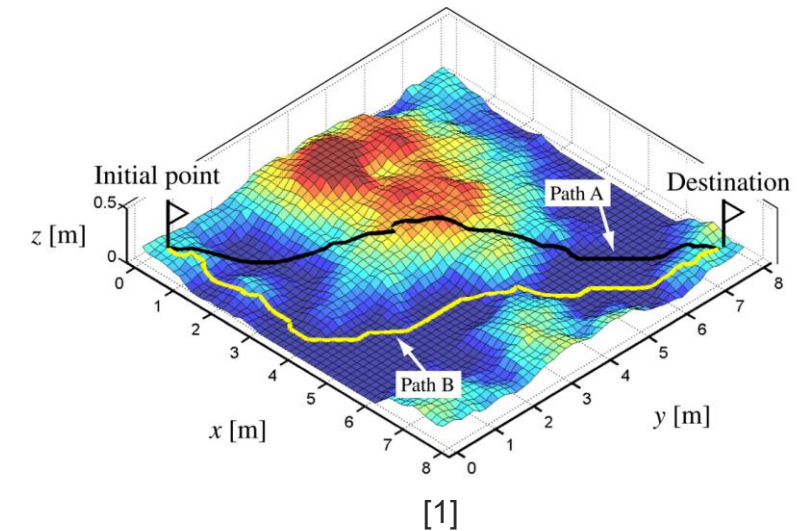
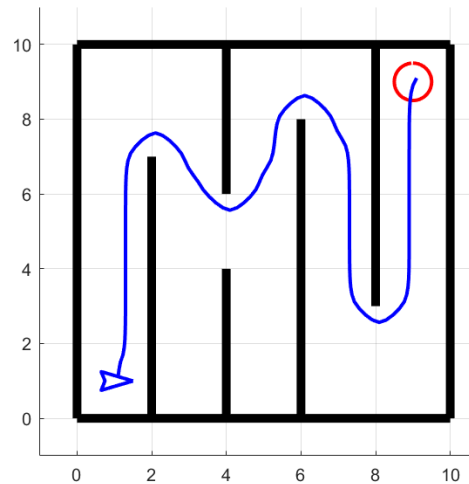
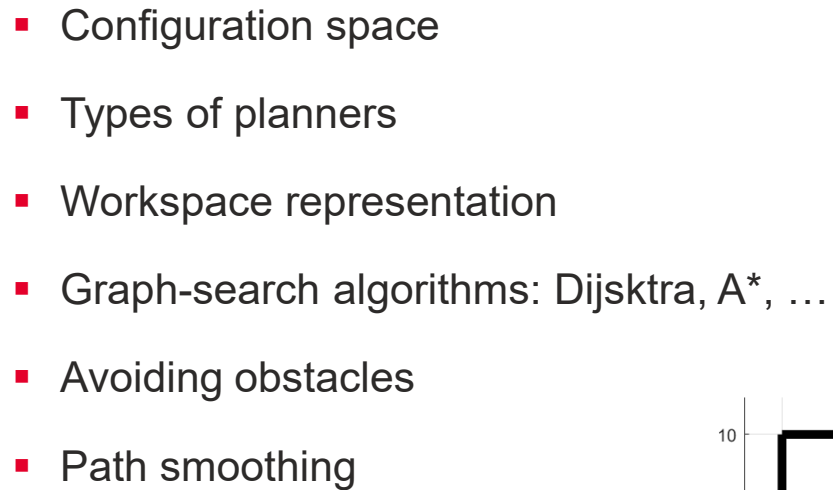


[3]

[1] GREWAL, M. S. and ANDREWS, A. P., 2010. Applications of Kalman Filtering in Aerospace 1960 to the Present [Historical Perspectives]. *IEEE Control Systems Magazine*. June 2010. Vol. 30, no. 3, p. 69–78. DOI [10.1109/MCS.2010.936465](https://doi.org/10.1109/MCS.2010.936465).

[2] Apollo command and service module, 2021. Wikipedia [online]. [Accessed 4 March 2021]. Available from: https://en.wikipedia.org/wiki/Apollo_command_and_service_module

[3] Implementations of Kalman Filter From Aerospace to Industry. P2 SMTP LIPI [online]. 2018 [cit. 2021-01-18]. Available at: <http://smtp.lipi.go.id/berita633-Implementations-of-Kalman-Filter-From-Aerospace-to-Industry.html>

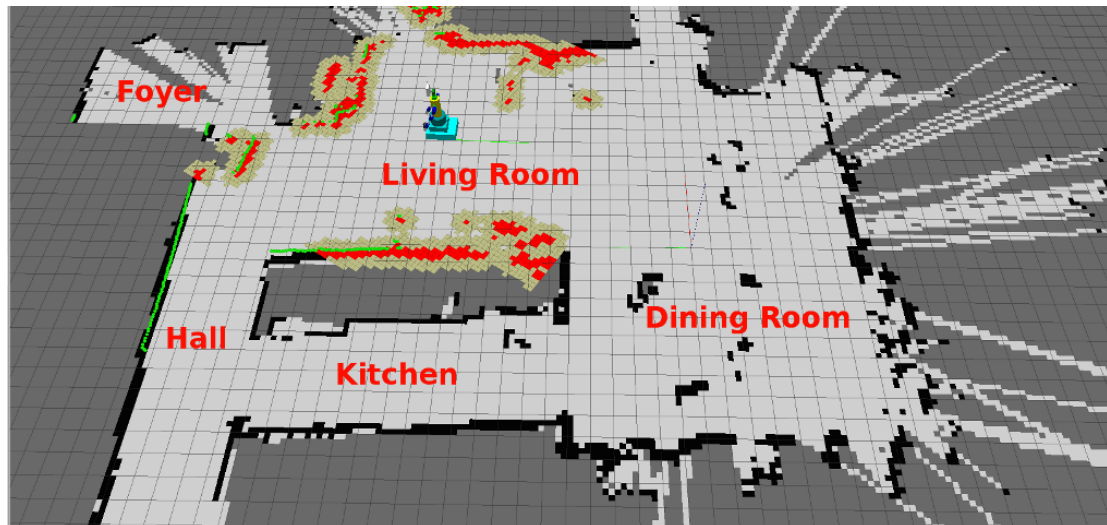


[2] Dijkstra's shortest path algorithm | Greedy Algo-7. In: *GeeksforGeeks* [online]. 2020 [cit. 2021-02-28]. Available at: <https://www.geeksforgeeks.org/dijkstras-shortest-path-algorithm-greedy-algo-7/>



EKF Slam

- **Simultaneous localization and mapping**
- Both tasks are addressed together
- Comprehensive algorithms
- Essential for indoor autonomous robots

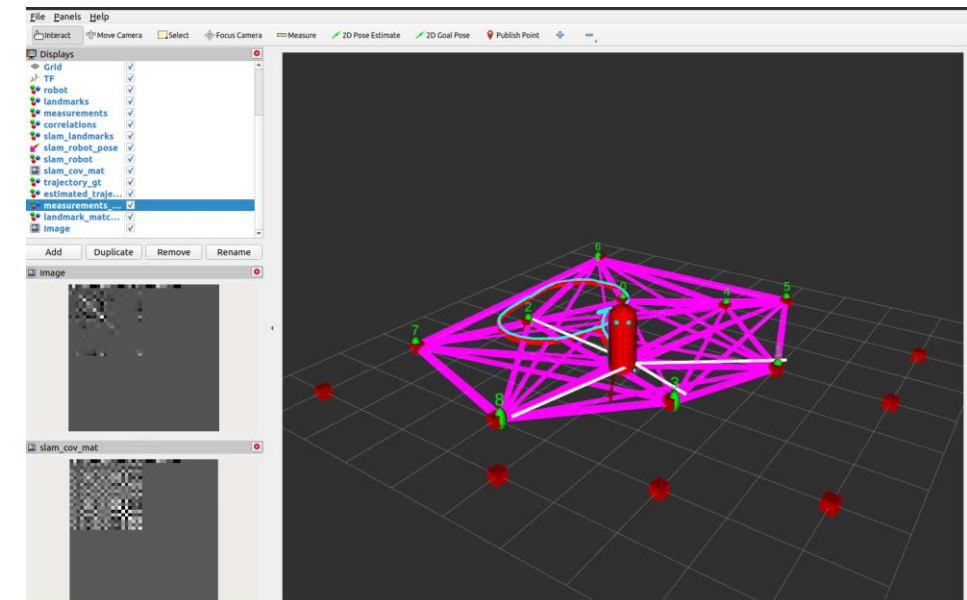


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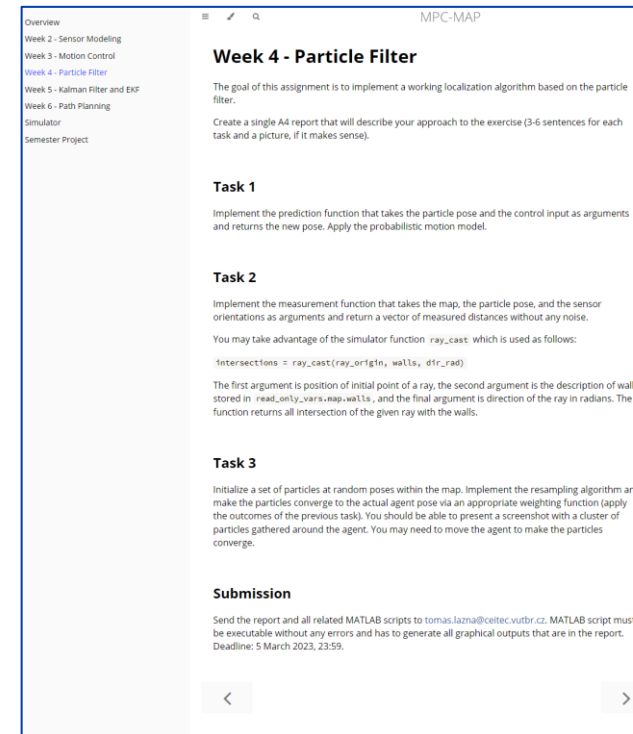
Particles-Based
SLAM

Graph-Based
SLAM

Kalman-Based
SLAM



- Weekly assignments in the weeks 2 to 6
- Task assignments available on the GitHub
- For each assignment you will prepare **short, single A4 report** describing your solution
- Each assignment task is documented via **a few sentences** and **image(s)** (if applicable)
- The deadline is next **Monday 23:59**
- During the week you can consult your solution with lecturers
- 0 to 10 points per assignment



Overview

- Week 2 - Sensor Modeling
- Week 3 - Motion Control
- Week 4 - Particle Filter
- Week 5 - Kalman Filter and EKF
- Week 6 - Path Planning
- Simulator
- Semester Project

Week 4 - Particle Filter

The goal of this assignment is to implement a working localization algorithm based on the particle filter.

Create a single A4 report that will describe your approach to the exercise (3-6 sentences for each task and a picture, if it makes sense).

Task 1

Implement the prediction function that takes the particle pose and the control input as arguments and returns the new pose. Apply the probabilistic motion model.

Task 2

Implement the measurement function that takes the map, the particle pose, and the sensor orientations as arguments and return a vector of measured distances without any noise.

You may take advantage of the simulator function `ray_cast` which is used as follows:

```
intersections = ray_cast(ray_origin, walls, dir_rad)
```

The first argument is position of initial point of a ray, the second argument is the description of walls stored in `read_only_vars.map.walls`, and the final argument is direction of the ray in radians. The function returns all intersection of the given ray with the walls.

Task 3

Initialize a set of particles at random poses within the map. Implement the resampling algorithm and make the particles converge to the actual agent pose via an appropriate weighing function (apply the outcomes of the previous task). You should be able to present a screenshot with a cluster of particles gathered around the agent. You may need to move the agent to make the particles converge.

Submission

Send the report and all related MATLAB scripts to `tomas.lazna@cetec.vutbr.cz`. MATLAB script must be executable without any errors and has to generate all graphical outputs that are in the report. Deadline: 5 March 2023, 23:59.

Assignment (GitHub)



BRNO UNIVERSITY OF TECHNOLOGY
FACULTY OF ELECTRICAL ENGINEERING
OF TECHNOLOGY AND COMMUNICATION

Robotics and AI

MPC-MAP Assignment No. 1 - Report

Author: Adam Ligocki
Date: 7st Feb 2022

Task 1

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Pellentesque faucibus, mi eu pellentesque venenatis, ligula leo tincidunt mauris, in tempus lectus erat eget purus. Fusce quis urna dolor. Phasellus tristique felis justo, vel consectetur magna luctus a. Nulla pharetra magna non pellentesque vestibulum.

Task 2

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Pellentesque faucibus, mi eu pellentesque venenatis, ligula leo tincidunt mauris, in tempus lectus erat eget purus. Fusce quis urna dolor. Phasellus tristique felis justo, vel consectetur magna luctus a. Nulla pharetra magna non pellentesque vestibulum.



Figure 1 - Lorem Ipsum

Task 3

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Pellentesque faucibus, mi eu pellentesque venenatis, ligula leo tincidunt mauris, in tempus lectus erat eget purus. Fusce quis urna dolor. Phasellus tristique felis justo, vel consectetur magna luctus a. Nulla pharetra magna non pellentesque vestibulum.

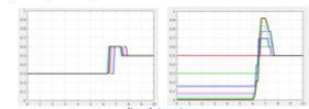
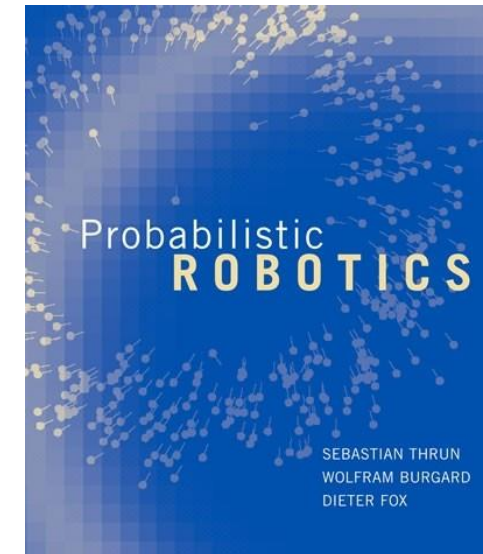


Figure 2 - Lorem Ipsum

Report (pdf)



- Lectures
- Labs / Consultations
- GitHub repository: <https://github.com/Robotics-BUT/MPC-MAP-Student>
- Sebastian Thrun: [Probabilistic Robotics](#)* (Book) [1]
- Sebastian Thrun: [Artificial Intelligence for Robotics](#) (Udacity course)



[1]

[1] THRUN, Sebastian, BURGARD, Wolfram and FOX, Dieter, 2005. Probabilistic Robotics. 1st edition. Cambridge, Mass: The MIT Press. ISBN 978-0-262-20162-9.

* An early book version is available online, the hardcover is available in the FEEC BUT library.



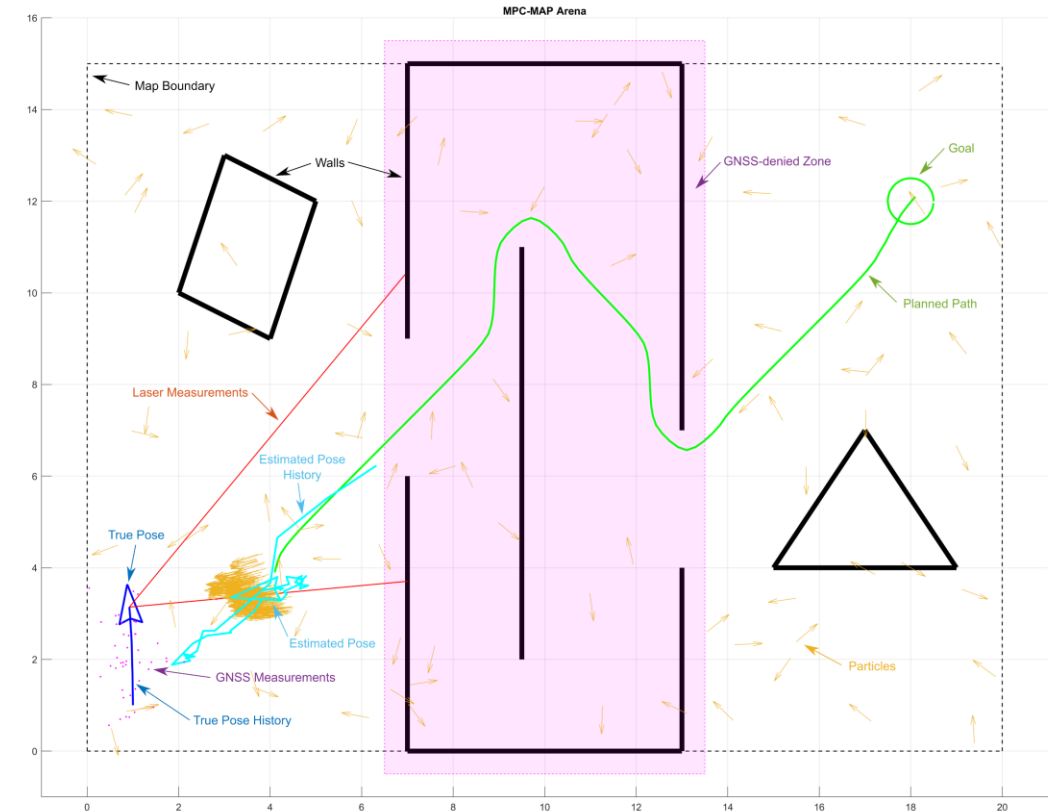
kalman ie estimator -

Simulator

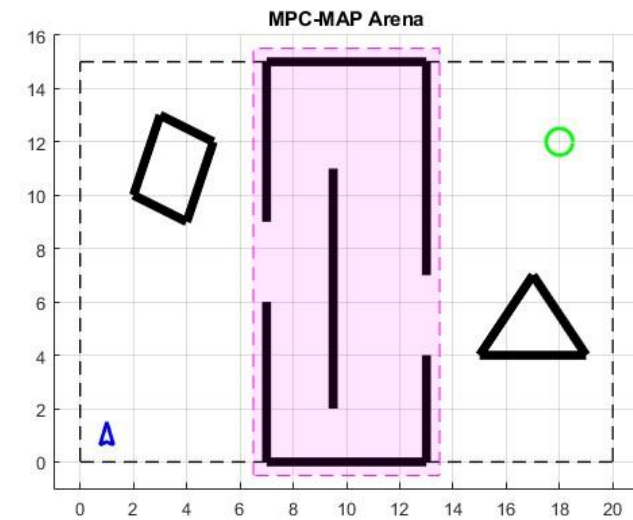
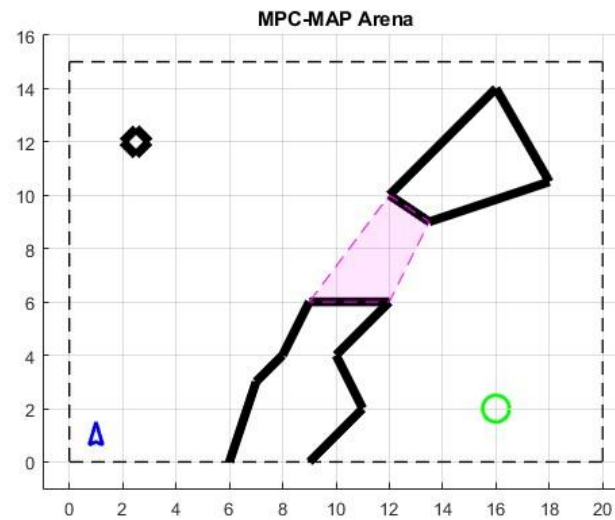
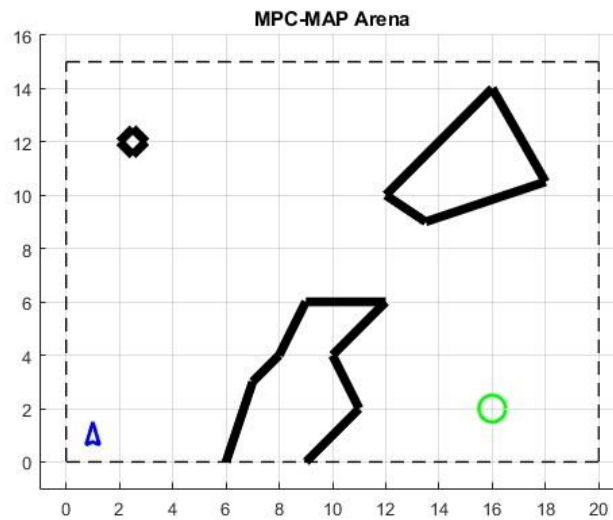
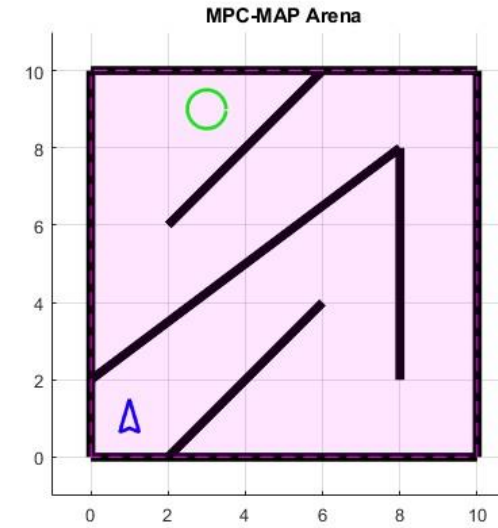
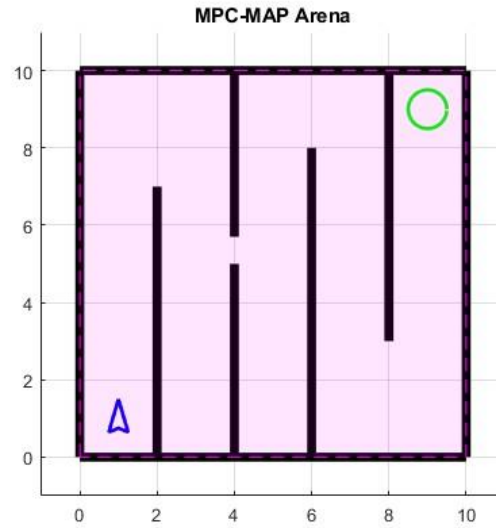
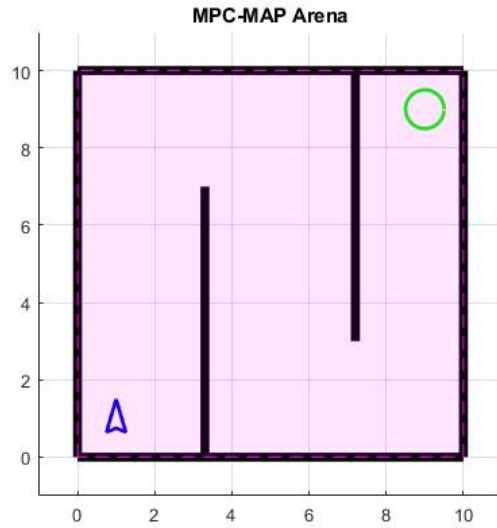
Lightweight, MATLAB-based robot simulator



- Both the **weekly assignments** and the **semestral project** are implemented in **MATLAB**
- **Custom-build**, lightweight **robot simulator**
- The robot (the blue arrow) is controlled via linear/angular speed
- Two sensors available: 8-way lidar, GNSS receiver
- The goal is to navigate the robot to the goal location
- Robot can not hit the walls and leave the map
- Numerous maps are available
- A simulation loop is prepared to be filled with **your custom functions** for **localization**, **path planning**, **motion control** etc.
- The simulator is available on GitHub



The simulation visualization



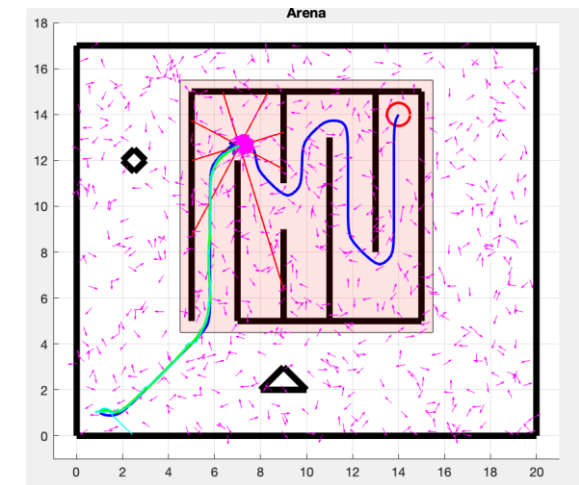
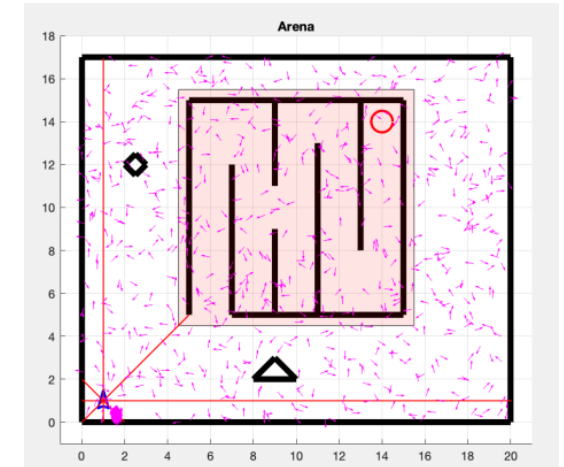


Semestral Project

The final project addressing a complete navigation problem

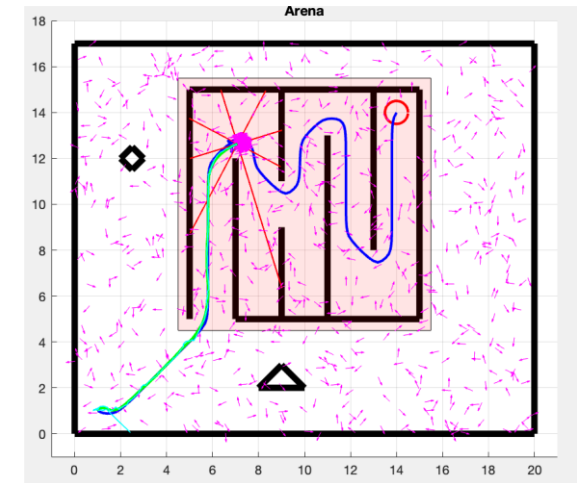
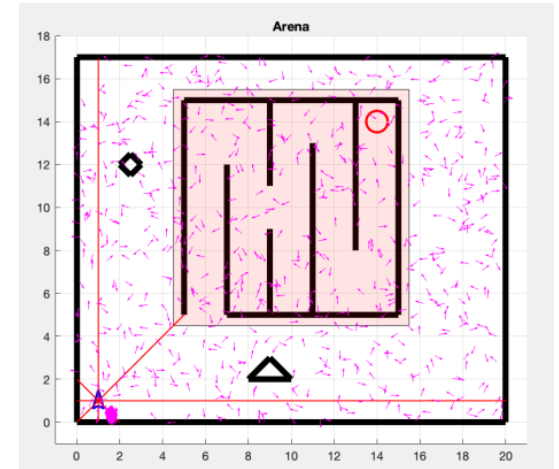


- The semestral project **connects all the functions prepared within weekly assignments together**
- The goal is to **navigate the robot to the goal** location:
 - With the minimal number of iterations
 - For diverse maps (incl. indoor, outdoor, combined)
 - For different start poses and goal locations
 - Without hitting walls and leaving the map
- A detailed semestral project assignment will be available on GitHub during the semester





- The project deadline: **13th week** of the semester, **Sunday 23:59**
- Your solutions will be tested and evaluated by repeated runs with different simulator settings (up to 100 runs)
- The solution (code quality, algorithms, ideas) will be evaluated as well
- Scoring – up to **50 points**:
 - Reaching goal probability: 0 to 15 points
 - Number of iterations: 0 to 15 points
 - Technical solution: 0 to 20 points



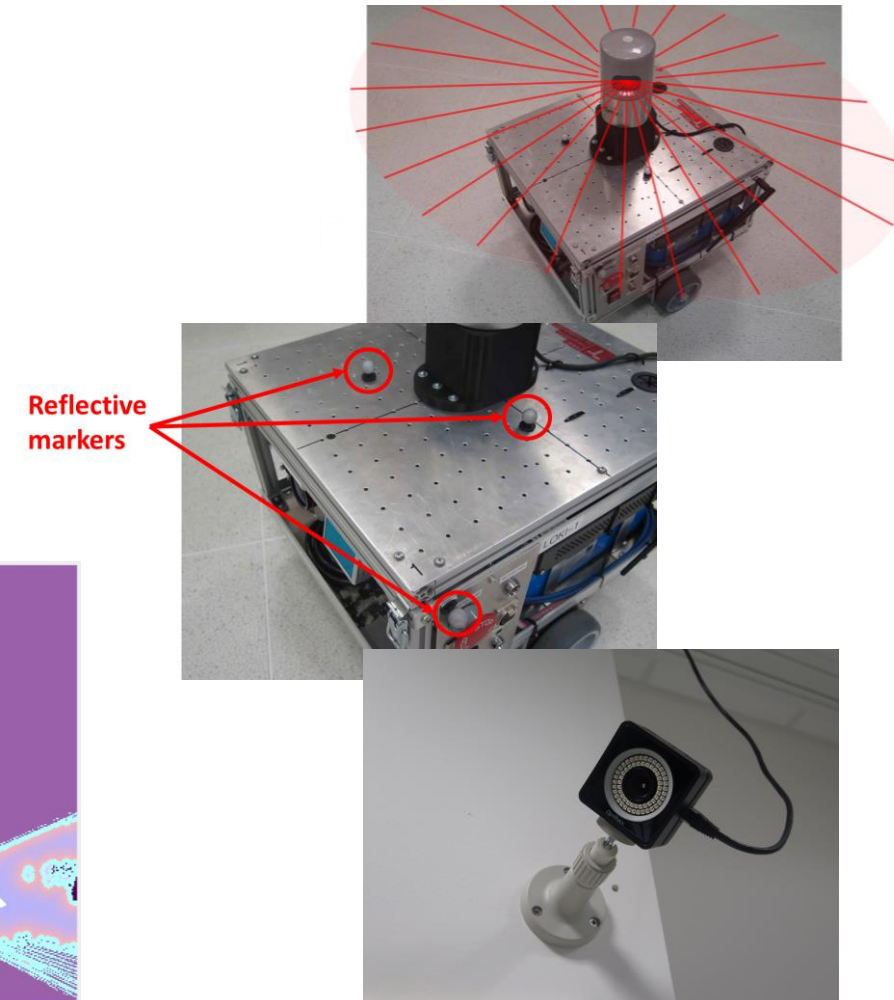
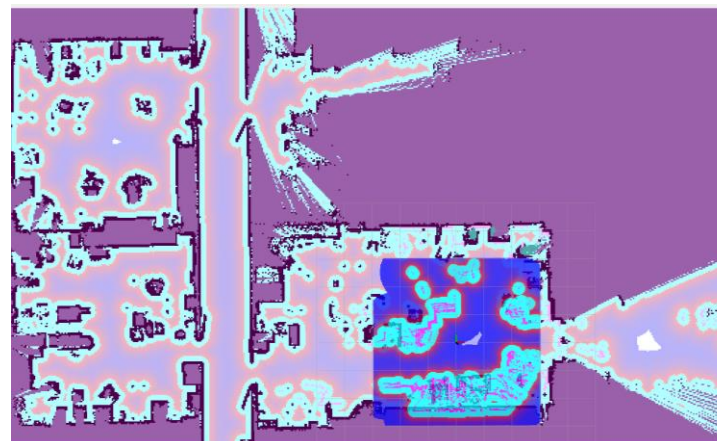
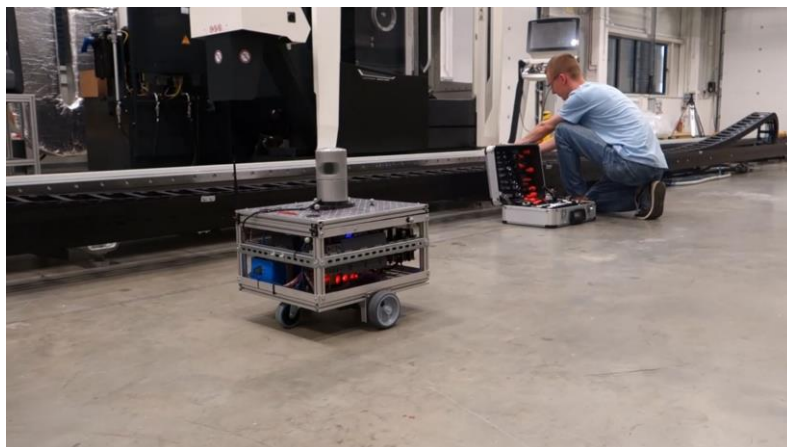


Live Demonstration

A live demo illustrating the concepts and algorithms



- **Loki** mobile robot
 - Custom build platform for the research and development
 - Differential drive
 - ROS2 based, incl. Nav2 stack
 - Sensors: encoders (odometry), 3D LiDAR, MoCap, IMU





Petr Gabrlik

gabrlik@vut.cz

Brno University of Technology
Faculty of Electrical Engineering and Communication
Department of Control and Instrumentation



Robotics and AI Research Group